

Week 0: Introduction

STAT 201A: Theory of Probability & Statistics

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Logistics

- **Lecture:** Tu/Th 11:00 a.m. – 12:20 p.m., Olmsted 1431
- **Discussion:** Th 10:00 – 10:50 a.m., Olmsted 1431
- **Instructor:** Zhe Fei (Olmsted 1344), zhe.fei@ucr.edu
- **Office hours:** Th 2:00 – 3:30 p.m., or by appointment
- **Course website:** Canvas (elearn.ucr.edu) for announcements, notes, homework, and exams; also feizhe.github.io/STAT201A

Description

4 units. Lecture, 3 hours; discussion, 1 hour. **Prerequisites:** MATH 010B and STAT 160C, each with a grade of C- or better; or consent of the instructor.

The STAT 201 sequence presents the techniques and basic results of probability and mathematical statistics at an advanced-calculus level. STAT 201A develops the probabilistic tools and language of mathematical statistics: probability models, random variables and random vectors, and their distributions.

Textbook

- George Casella and Roger L. Berger, *Statistical Inference*, 2nd Edition, Duxbury (2002).
- [Online version](#)

We cover **Chapters 1–4**. Reading the relevant sections **before** lecture is strongly recommended.

Course Goals

By the end of the quarter, you should be able to:

1. **Speak the language of probability precisely.** Sample spaces, events, σ -algebras, and probability functions, stated as definitions and used correctly.
2. **Compute probabilities.** Use counting, conditioning, independence, and Bayes' rule, and know which tool fits which problem.
3. **Work with random variables.** Move between cdfs, pmfs/pdfs, and transformations $Y = g(X)$.
4. **Compute and reason with expectations.** Means, variances, moments, and moment generating functions, and use the mgf to identify distributions.
5. **Know the common families.** Binomial, Poisson, geometric/negative binomial, hypergeometric, uniform, normal, gamma/exponential, χ^2 , beta, Cauchy: where each comes from and how they are related.
6. **Handle several random variables at once.** Joint, marginal, and conditional distributions; independence; bivariate transformations; hierarchical models; covariance and correlation; the multinomial and bivariate normal distributions.
7. **Write clear, rigorous derivations and proofs.** This is the main skill the course trains, and it is what the rest of the 201 sequence and the qualifying exam build on.

How this course differs from STAT 160

	Undergraduate probability	STAT 201A
Goal	Get the right number	Justify <i>why</i> the number is right
Definitions	Intuitive	Precise (axioms, set theory)
Typical question	"Find $P(A)$ "	"Show that $P(A \cup B) \leq P(A) + P(B)$ for any P satisfying the axioms"
Tools	Formulas	Calculus, series, integration, careful logic

Expectations

Work load

Plan on **3–4 hours of work outside class for each hour of lecture**. Most of that time should go to working problems with pencil and paper. Reading a solution is not the same as being able to produce one.

How to study for a theory course

- **Read before lecture, re-derive after.** After each lecture, close the notes and reproduce the key derivations yourself.
- **Learn the definitions exactly.** Most mistakes in proofs come from a definition that was only half remembered.
- **Write complete arguments.** Every step should follow from a definition, an axiom, or a result proved earlier. “It is obvious” is not a step.
- **Try examples and counterexamples.** Before proving a claim, test it on a small case. Before accepting a claim, look for a way it could fail.
- **Come to office hours and discussion.** Bring your partial attempts. They are much more useful than a blank page.

Attendance

Attendance at lectures and discussions is expected and counts toward your grade. If you cannot attend because of medical or other circumstances, contact the instructor as early as possible.

Assessment

Grading

Component	Weight	Date
Homework (biweekly)	20%	posted on Canvas
Quizzes	10%	during Discussions
Midterm	25%	Thu. Nov. 5
Final	35%	Thu. Dec. 10
Attendance	10%	

Homework

- Assigned **bi-weekly** and submitted on Canvas.
- **No late homework** is accepted without prior arrangement with the instructor.
- Typeset solutions (LaTeX, or Markdown with math) are encouraged. Neat, scanned handwriting is also acceptable.
- You are welcome, and encouraged, to discuss problems with classmates, but **the solution you submit must be written by you**, in your own words. Copying another student’s work, or an AI-generated solution, is an academic integrity violation (see below).

Exams

- The midterm and final are **closed-book, in-person** exams.
 - There are no make-up exams without an officially excused absence arranged **in advance**.
 - Using disallowed materials during an exam (notes, devices, AI tools, communication with others) results in a 0 and will be reported to [Student Conduct & Academic Integrity Programs \(SCAIP\)](#).
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Tentative Topics

See the [weekly schedule](#) for dates.

1. **Probability theory** (Ch. 1): set theory; axioms of probability; counting; conditional probability and Bayes' rule; independence; random variables; cdfs; pmfs and pdfs.
 2. **Transformations and expectations** (Ch. 2): distributions of $g(X)$; expected values; moments and moment generating functions.
 3. **Common families of distributions** (Ch. 3): discrete and continuous families; exponential families; location–scale families; probability inequalities and identities.
 4. **Multiple random variables** (Ch. 4): joint and marginal distributions; conditional distributions and independence; bivariate transformations; hierarchical models and mixtures; covariance and correlation; multivariate distributions; inequalities (e.g., Cauchy–Schwarz, Jensen).
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Use of AI

AI tools, especially large language models (LLMs) such as ChatGPT, Claude, and Gemini, are now part of how mathematics and statistics get done. Used well, they can speed up your learning a lot. Used badly, they replace the exact thinking this course is meant to build.

A probability connection

An LLM is a probability model. Given the text so far (the *context*), it defines a conditional distribution over the next token w :

$$P(w \mid \text{context}) = \frac{\exp(z_w/T)}{\sum_v \exp(z_v/T)},$$

where z_v are scores computed by a neural network and $T > 0$ is the *temperature*. The response is produced by **sampling** from this distribution one token at a time. Some consequences:

- The same question can produce different answers. The output is a random variable.
- A fluent, confident answer is a *likely* sequence of words, not necessarily a *correct* argument.

- As $T \rightarrow 0$, the distribution concentrates on the most likely token (assuming a unique largest score). As T grows, it approaches uniform. (Exercise: prove both claims once we have covered limits of probabilities.)

By the end of this course you will have all the tools to read that formula precisely: sample spaces, conditional probability, and discrete distributions.

Recommended uses

- **Concept tutor.** “Explain the difference between a pmf and a cdf, with an example where the cdf has jumps.”
- **Socratic partner.** “I am trying to prove Boole’s inequality. Don’t give me the proof. Ask me questions that lead me to it.”
- **Checking your own work.** Write your derivation first, then ask the AI to find gaps or errors in *your* argument.
- **Extra practice.** “Give me three problems like Casella & Berger 1.13, without solutions.”
- **LaTeX and typesetting.** Formatting equations, aligning derivations, fixing compile errors.
- **Numerical sanity checks.** Ask for a short simulation that checks a probability you derived by hand.

Not recommended / not allowed

- **Submitting AI-generated solutions** to homework as your own work.
- **Any AI use during quizzes or exams.**
- Asking the AI to “solve Problem 1.x” before you have seriously attempted it. You skip exactly the struggle that builds the skill, and exams have no AI.

Always verify

LLMs regularly make **confident mistakes** in exactly the areas this course covers:

- **Counting problems:** double counting, and mixing up ordered and unordered selections, or sampling with and without replacement.
- **Conditional probability:** confusing $P(A \mid B)$ with $P(B \mid A)$.
- **Proofs:** circular reasoning, steps that are “clear” but false, and conditions that are assumed without being checked (e.g., interchanging limits, sums, and integrals).
- **Citations:** made-up theorem numbers, page references, and papers.

Rule of thumb: if you could not defend every line of an AI answer at the whiteboard, you do not understand it yet, and you should not rely on it.

AI as a helper, never as a replacement. You ask the questions, you do the reasoning, and you check every answer.

Preview: Probability and Uncertainty

Probability is the mathematical language of **uncertainty**. Statistics uses it in reverse: we observe data and reason back to the mechanism that produced it. In STAT 201A we build that language carefully, starting from sets.

Experiments, outcomes, and sample spaces (C&B §1.1)

Definition. The set S of all possible outcomes of a particular experiment is called the **sample space** for the experiment.

Examples.

1. Toss a coin once: $S = \{H, T\}$.
2. Toss a coin twice: $S = \{HH, HT, TH, TT\}$.
3. Count the students who arrive late today: $S = \{0, 1, 2, \dots\}$ (**countable**).
4. Measure the time until the next email arrives: $S = (0, \infty)$ (**uncountable**).

Definition. An **event** is any collection of possible outcomes of an experiment, i.e., any subset of S (including S itself).

For example, in Example 2 the event “at least one head” is $A = \{HH, HT, TH\} \subset S$.

Question for next time. Is every subset of $S = (0, \infty)$ an event we can assign a probability to? The answer, *not necessarily*, is why we will need σ -algebras in Week 1.

Quotes

“All models are wrong, but some are useful.” (George Box)

“You can, for example, never foretell what any one man will do, but you can say with precision what an average number will be up to. Individuals vary, but percentages remain constant. So says the statistician.” (Sherlock Holmes, *The Sign of Four*)

“Probability theory is nothing but common sense reduced to calculation.” (Pierre-Simon Laplace)